MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG1DSCBIF100 – Fundamental IT for Bioinformatics

Duration: 2 hrs Maximum Marks: 50

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions
Answer any 5 out of 7 Questions
Each question carries 2 mark

1. Name an example of an input device. [K][CO1]
2. Recognize the role of the CPU in a computer. [K][CO1]
3. Recognize the purpose of a computer network [K][CO2]
4. Outline the use of the `<img>` tag in HTML. [K][CO3]
5. Recognize the function of CSS in a web page. [K][CO3]
6. Demonstrate the use of an if-else statement in JavaScript. [U][CO3]
7. Identify basic terminologies in Bioinformatics [K] [CO4]

[2 x 5 = 10]

Part B

Short Essay Type Questions
Answer any 4 out of 6 Questions
Each question carries 5 marks

8. Compare primary memory and secondary storage in terms of speed, volatility, and use cases. [U][CO1]
9. Differentiate between compilers and interpreters and explain how each processes code. [U][CO3]
10. Identify common HTML elements for text formatting and demonstrate their uses with examples. [K][CO3]
11. Describe how JavaScript uses conditional statements for decision-making. [S][CO3]
12. Summarize history of Bioinformatics. [U] [CO4]
13. Comprehend Human Genome Project. [U] [CO4]

[5 x 4 = 20]

Part C

Long Essay Type Questions
Answer any 2 out of 4 Questions
Each question carries 10 marks

14. Explain the purpose of network protocols and describe examples like HTTP, HTTPS, FTP, and SMTP. [U][CO2]
15. List and explain the main tags used in HTML for creating headings, paragraphs, and links. Describe their functions. [K][CO3]
16. Explain basic terminologies and important areas of Bioinformatics. [U] [CO4]
17. Discuss any 4 applications of Bioinformatics. [U] [CO4]

[10 x 2 = 20]